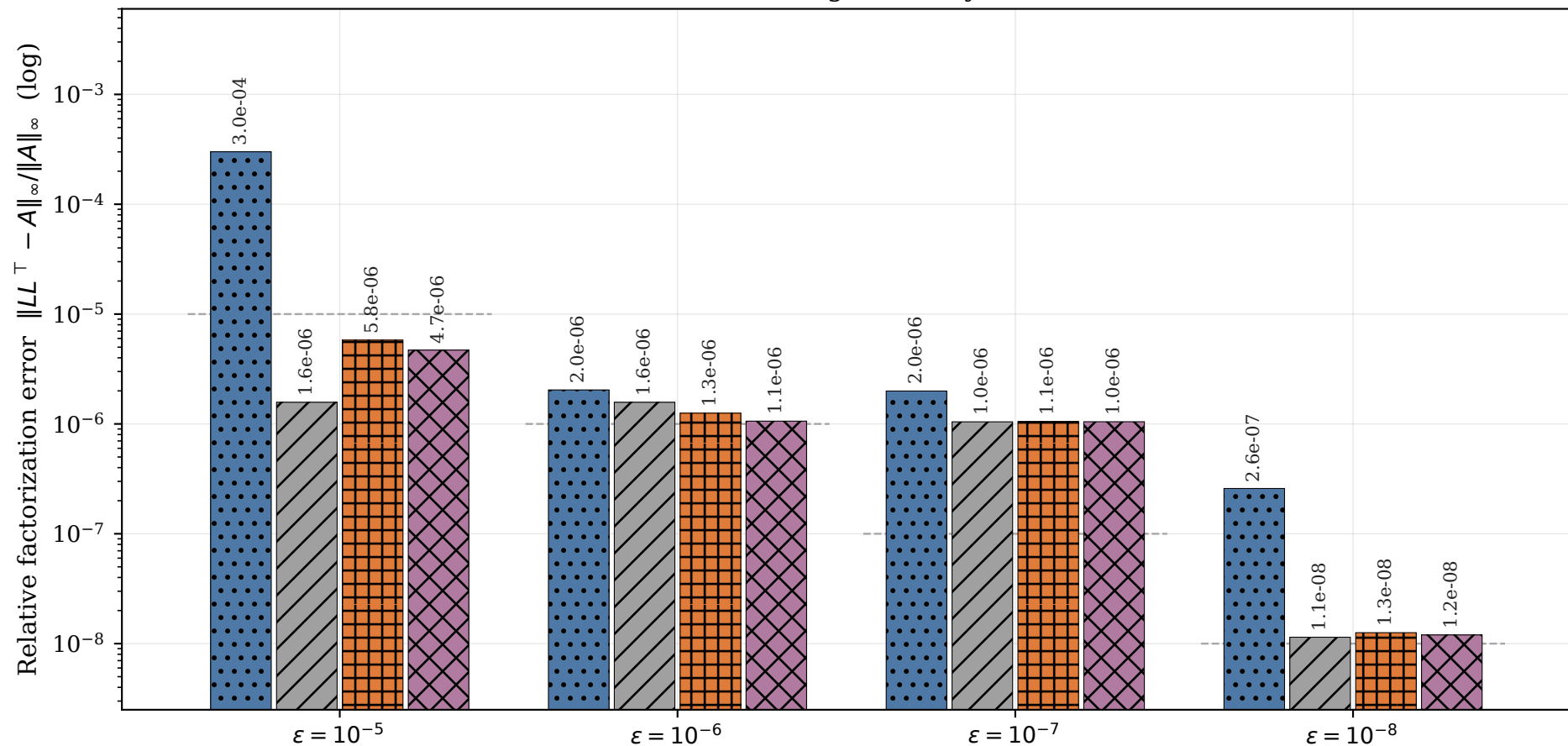


Ladder mode at tile + vec1D-32 granularity, N = 40960, NB = 4096



Bar order within each cluster (left -> right) . grey dashed = source  $\epsilon$

- Plain FP8 baseline (no MX; subnormal-allowed)
- IEEE ladder (FP8 E4M3 -> FP16 -> FP32 -> FP64; no MX)
- Full ladder tile (MXFP4 -> MXFP8 -> MXFP16 -> FP32 -> FP64)
- Full ladder vec1D-32 (MXFP4 -> MXFP8 -> MXFP16 -> FP32 -> FP64)